1. Project Title

Electric Vehicle Sales and Market Demand Analysis Using Business Intelligence

2. Business Problem Statement

The rapid growth of the electric vehicle (EV) industry has created a need for businesses to understand the factors influencing EV sales performance. Key variables such as charging infrastructure availability, fuel prices, and consumer demand trends play a significant role in market adoption. However, organizations often lack structured insights to evaluate these relationships and make data-driven strategic decisions.

This project aims to analyze electric vehicle sales data, charging station infrastructure, and fuel price trends to identify patterns, correlations, and growth opportunities that can support business planning and market expansion strategies.

3. Project Objectives

- *Analyze EV sales performance across time and markets*
- *Identify factors influencing EV adoption*
- *Measure growth trends and sales momentum*
- *Evaluate relationship between fuel prices and EV sales*
- *Assess impact of charging infrastructure on vehicle sales*
- *Provide data-driven business recommendations*

4. Stakeholders

Primary Stakeholders:

- *Sales Manager*
- *Business Analyst*
- *Marketing Team*
- *Operations Team*

Secondary Stakeholders:

- *Strategy Team*
- *Executive Management*
- *Product Planning Team*

5. Business Requirements.

BR-1 — Sales Performance Monitoring

The system shall provide visibility into monthly electric vehicle sales performance across different countries and brands.

BR-2 — Market Trend Analysis

The system shall enable analysis of EV sales trends over time to identify growth patterns and seasonal demand changes.

BR-3 — Infrastructure Impact Analysis

The system shall allow business users to evaluate how charging infrastructure availability affects EV sales performance.

BR-4 — Fuel Price Impact Analysis

The system shall provide insights into the relationship between gasoline prices and EV adoption rates.

BR-5 — Brand Performance Tracking

The system shall identify top-performing EV brands based on sales volume.

BR-6 — Growth and Momentum Tracking

The system shall track year-over-year growth and sales momentum to support strategic planning.

BR-7 — Demand Trend Monitoring

The system shall analyze market demand trends using EV-related search trend data.

6. Functional Requirements (FRD Section).

Data Processing

FR-1

The system shall import EV sales data from the cleaned dataset.

FR-2

The system shall calculate total units sold by month.

FR-3

The system shall calculate sales growth percentage.

FR-4

The system shall calculate rolling average sales.

FR-5

The system shall calculate lag sales values.

Analysis Features

FR-6
The system shall perform correlation analysis between:

- *Gasoline price and EV sales*
- *Charging stations and EV sales*

FR-7
The system shall calculate brand-level sales totals.

FR-8
The system shall identify the highest sales brand.

FR-9
The system shall calculate growth momentum trends.

Reporting and Visualization

FR-10
The system shall generate monthly sales trend charts.

FR-11
The system shall generate correlation charts.

FR-12
The system shall generate growth trend visualizations.

FR-13
The system shall display sales performance dashboards.

7. Non-Functional Requirements

Performance

The system should generate reports within 5 seconds.

Accuracy

The system should ensure calculations are accurate and consistent.

Usability

The dashboard should be easy to understand for business users.

Reliability

The system should handle large datasets without failure.

Security

Access to the dataset should be restricted to authorized users.

8. Key Performance Indicators (KPIs)

Sales KPIs

- *Total Units Sold*
- *Monthly Sales Growth*
- *Rolling Average Sales*
- *Year-over-Year Growth*

Market KPIs

- *Sales Growth Momentum*
- *Demand Trend Index*
- *EV Adoption Rate*

Infrastructure KPIs

- *Total Charging Stations*
- *Charging Station Growth Rate*
- *Infrastructure Coverage*

Financial KPIs

- *Average Gasoline Price*
- *Fuel Price Trend*
- *Price Impact on Sales*

9. Data Sources

Primary Dataset

strictly_cleaned_ev_dataset.csv

Contains:

- *EV sales data*
- *Charging station infrastructure*
- *Fuel price data*

- *Brand performance*
- *Sales growth metrics*

Analysis Dataset

ev_analysis.xlsx

Contains:

- *Growth analysis*
- *Correlation analysis*
- *Sales trend analysis*
- *Brand performance analysis*

Demand Dataset

ev_trends_monthly.xlsx

Contains:

- *EV search trend data*
- *Market demand indicators*
- *Monthly trend scores*

10. Data Analysis Summary

- *The analysis focused on identifying patterns and relationships between electric vehicle sales, charging infrastructure, fuel prices, and market demand trends. Correlation analysis was performed to determine how gasoline prices and charging station availability influence EV adoption. Sales growth momentum and rolling averages were calculated to understand long-term performance trends. Brand-level performance analysis was conducted to identify leading manufacturers in the EV market.*
- *The analysis also incorporated market demand data using EV-related search trends to evaluate consumer interest over time. These insights support data-driven decision-making for market expansion, infrastructure planning, and sales strategy development.*

11. Key Insight

Insight 1 — EV Sales Show Consistent Upward Growth

The analysis of monthly units sold indicates a steady increase in electric vehicle sales over time. This suggests growing market adoption and increasing consumer acceptance of electric vehicles.

Business Meaning:
The EV market is expanding, and companies should prepare for increased demand by scaling production and distribution.

Insight 2 — Charging Infrastructure Positively Influences EV Sales

Correlation analysis shows a strong positive relationship between the number of charging stations and EV sales volume. Regions with more charging stations tend to have higher vehicle sales.

Business Meaning:
Investment in charging infrastructure directly supports EV adoption and sales growth.

Insight 3 — Rising Fuel Prices Encourage EV Adoption

The analysis indicates that higher gasoline prices are associated with increased EV sales. As fuel costs rise, consumers are more likely to consider electric vehicles as a cost-saving alternative.

Business Meaning:
Fuel price trends can be used as a predictive indicator for EV demand.

Insight 4 — Certain Brands Dominate the Market

Brand-level analysis shows that a small number of EV manufacturers contribute a large portion of total sales. These brands consistently outperform others in market share

Business Meaning:
Businesses should monitor competitor performance and benchmark strategies against leading brands.

Insight 5 — Sales Momentum Indicates Market Expansion

Growth momentum analysis shows sustained positive growth over multiple periods, indicating strong market expansion and increasing demand.

Business Meaning:
The EV industry is in a growth phase, presenting opportunities for investment and expansion.

Insight 6 — Search Trends Reflect Consumer Interest

The EV search trend dataset shows increasing search activity for electric vehicles over time, indicating rising consumer awareness and interest.

Business Meaning:
Search trends can be used as an early signal for future sales demand.

12. Business Recommendations

Recommendation 1 — Expand Charging Infrastructure

Businesses and governments should prioritize building additional charging stations in regions with growing EV demand.

Expected Impact:

- *Increased EV adoption*
- *Improved customer convenience*
- *Higher sales growth*

Recommendation 2 — Monitor Fuel Price Trends for Demand Forecasting

Organizations should track gasoline price changes to anticipate shifts in EV demand and adjust production or inventory accordingly.

Expected Impact:

- *Better demand forecasting*
- *Reduced inventory shortages*
- *Improved operational planning*

Recommendation 3 — Focus Marketing Efforts in High-Growth Regions

Marketing campaigns should target regions showing strong sales growth and increasing consumer interest.

Expected Impact:

- *Higher conversion rates*
- *Increased market penetration*
- *Improved return on marketing investment*

Recommendation 4 — Strengthen Partnerships with Top-Performing Brands

Businesses should collaborate with leading EV manufacturers to leverage their market presence and brand recognition.

Expected Impact:

- *Increased sales opportunities*
- *Competitive advantage*
- *Stronger market positioning*

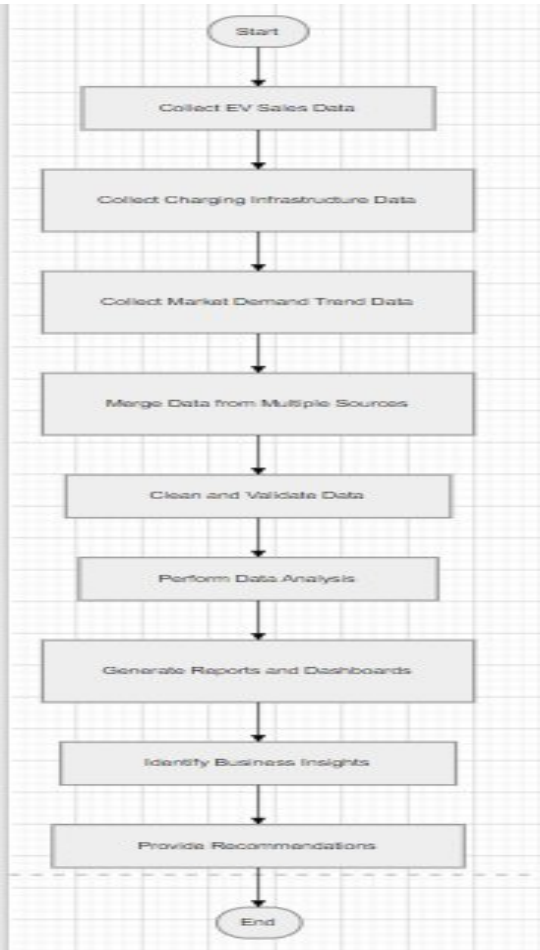
Recommendation 5 — Use Market Trends for Strategic Planning

Organizations should integrate sales trends, infrastructure data, and search trends into long-term planning and investment decisions.

Expected Impact:

- *Data-driven decision-making*
- *Improved strategic planning*
- *Sustainable business growth*

13. PROCESS FLOW DIAGRAM

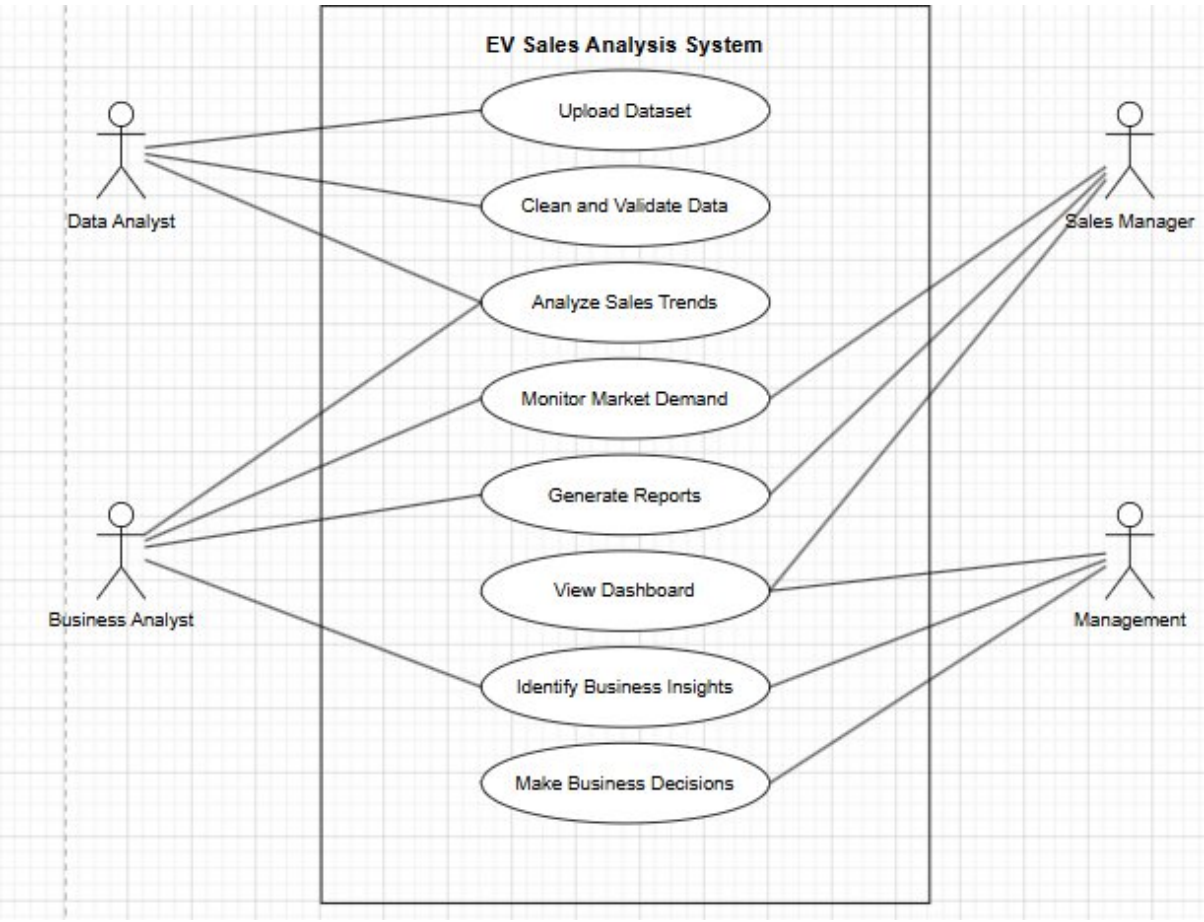


Process Flow Description

The process flow diagram illustrates the workflow of analyzing electric vehicle sales and market demand data. The process begins with collecting data from multiple sources, including EV sales records, charging infrastructure information, and market trend datasets. The collected data is merged, cleaned, and validated to ensure accuracy and consistency. The validated data is then analyzed to identify patterns, trends, and correlations. Reports and dashboards are generated to visualize key metrics and performance indicators. Finally, business insights and recommendations are provided to support strategic decision-making and market planning.

14. USE CASE DIAGRAM

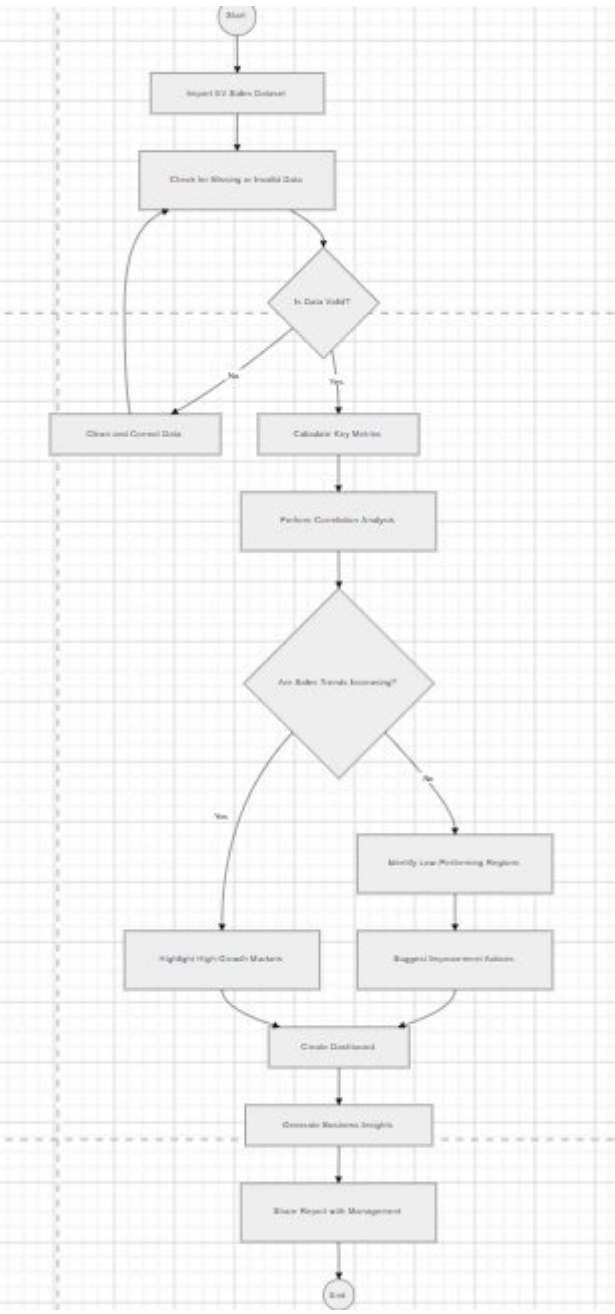
Use Case Diagram: Electric Vehicle Sales and Market Analysis System



The use case diagram illustrates how different stakeholders interact with the electric vehicle sales analysis system. The Data Analyst is responsible for uploading, cleaning, and validating the dataset. The Business Analyst performs data analysis, generates reports, and identifies key business insights. The Sales Manager uses the dashboard to monitor sales performance and market trends. Management reviews the insights and makes strategic business decisions based on the analysis results.

15. ACTIVITY DIAGRAM

Activity Diagram: Electric Vehicle Sales and Market Analysis Workflow



*The activity diagram represents the detailed workflow of analyzing electric vehicle sales and market data. The process begins with importing and validating the dataset to ensure data quality. If errors are detected, the data is cleaned and revalidated. Once the data is confirmed to be accurate, key metrics such as sales growth and infrastructure impact are calculated. Correlation analysis is performed to understand relationships between variables. Based on the analysis results, the system identifies high-growth or low-performing markets and generates insights. Finally, dashboards and reports are created and shared with management to support strategic decision-making***.**

16. Functional Requirements Document (FRD)

Project Name

Electric Vehicle Sales and Market Demand Analysis.

FR ID	Functional Requirement	Description	Priority	Stakeholder
FR-1	Import Dataset	The system shall allow users to upload EV sales and trend datasets	High	Data Analyst
FR-2	Validate Data	The system shall check for missing or invalid data values	High	Business Analyst
FR-3	Clean Data	The system shall allow correction of incomplete or incorrect data	High	Data Analyst
FR-4	Calculate KPIs	The system shall calculate total sales, growth rate, and infrastructure metrics	High	Management
FR-5	Perform Trend Analysis	The system shall analyze sales trends over time	High	Business Analyst
FR-6	Perform Correlation Analysis	The system shall analyze relationships between fuel price, charging stations, and sales	Medium	Business Analyst
FR-7	Generate Dashboard	The system shall display interactive dashboards with charts and KPIs	High	Sales Manager
FR-8	Generate Reports	The system shall produce summary reports of analysis results	High	Management
FR-9	Identify Top Brands	The system shall identify brands with the highest sales performance	Medium	Marketing Team
FR-10	Share Insights	The system shall allow users to share insights with stakeholders	Medium	Business Analyst

Non-Functional Requirements (FRD Section)

Performance

The system should generate reports within 5 seconds.

Accuracy

The system should ensure correct calculations and reliable outputs.

Usability

The dashboard should be easy to understand for business users.

Reliability : The system should handle datasets with large volumes of records**.**

17. USER STORIES

User Story ID	User Story	Acceptance Criteria	Priority
US-1	As a Data Analyst, I want to upload the EV dataset so that I can analyze sales data	Dataset uploads successfully	High
US-2	As a Business Analyst, I want to validate data so that I can ensure accuracy before analysis	Missing values are detected	High
US-3	As a Business Analyst, I want to calculate KPIs so that I can measure business performance	KPIs display correctly	High
US-4	As a Sales Manager, I want to view dashboards so that I can monitor sales performance	Dashboard loads successfully	High
US-5	As a Manager, I want to see monthly sales trends so that I can make strategic decisions	Trend charts are visible	High
US-6	As a Business Analyst, I want to analyze correlations so that I can understand factors influencing sales	Correlation results are displayed	Medium
US-7	As a Manager, I want to receive reports so that I can review business performance	Reports generate successfully	Medium
US-8	As a Business Analyst, I want to identify top-performing brands so that I can recommend business strategies	Top brand is displayed	Medium

18. User Acceptance Testing (UAT)

UAT Scope

The testing covers:

- *Dataset upload and validation*
- *Data cleaning and processing*
- *KPI calculation*
- *Trend and correlation analysis*
- *Dashboard visualization*
- *Report generation*
- *Business insights delivery*

UAT ENVIRONMENT

Component	Details
Tool Used	Microsoft Excel / Google Sheets
Dataset	EV Sales Dataset
Users	Business Analyst, Data Analyst, Manager
Testing Type	Manual Testing
Data Size	Historical EV sales records

UAT TESTCASES

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
UAT-1	Upload Dataset	Upload EV dataset into the system	Dataset uploads successfully	Pass
UAT-2	Validate Data	Check dataset for missing values	Missing values are identified	Pass
UAT-3	Clean Data	Remove or correct invalid data	Data becomes consistent and accurate	Pass
UAT-4	Calculate KPIs	Generate total sales and growth metrics	KPIs display correctly	Pass
UAT-5	Generate Dashboard	Create charts and visualizations	Dashboard displays correctly	Pass
UAT-6	Analyze Trends	Generate sales trend analysis	Trend patterns are visible	Pass
UAT-7	Perform Correlation Analysis	Compare fuel price and EV sales	Correlation results are displayed	Pass
UAT-8	Generate Report	Export analysis summary	Report is generated successfully	Pass

UAT Acceptance Criteria

The system will be accepted if:

- Data is uploaded without errors
- Calculations are accurate
- Dashboards display correct values
- Reports generate successfully
- Insights are understandable for business users

UAT SIGNOFF

Role	Name	Signature	Date
Business Analyst	_____	_____	_____
Project Manager	_____	_____	_____
Stakeholder	_____	_____	_____

19. BUSINESS REQUIREMENTS DOCUMENT(BRD) SUMMARY

Project Name

Electric Vehicle Sales and Market Demand Analysis

Purpose

*The purpose of this project is to analyze electric vehicle sales data, charging infrastructure availability, and market demand trends to identify key factors influencing EV adoption and sales performance. The analysis supports data-driven decision-making for business strategy, infrastructure planning, and market expansion***.**

Scope

This project includes:

- *Analysis of electric vehicle sales data*
- *Evaluation of charging infrastructure impact*
- *Assessment of fuel price influence on EV adoption*
- *Identification of sales growth trends*
- *Visualization of performance using dashboards*
- *Generation of actionable business insights*

This project does not include:

- *Real-time data integration*
- *Predictive machine learning models*
- *Automated reporting systems*

Business Objectives

- *Improve visibility into EV sales performance*
- *Identify factors affecting EV adoption*
- *Monitor market demand trends*
- *Support strategic planning and decision-making*
- *Optimize resource allocation and infrastructure investment*

Stakeholders

Primary Stakeholders:

- *Business Analyst*
- *Sales Manager*
- *Operations Manager*
- *Executive Management*
-

Secondary Stakeholders:

- *Marketing Team*
- *Strategy Team*
- *Product Planning Team*

20. Project Deliverables

Deliverables

The following deliverables were produced during the project:

- *Cleaned EV Dataset*
- *Data Analysis Reports*
- *Business Dashboard*
- *Key Performance Indicators (KPIs)*
- *Business Insights*
- *Business Recommendations*
- *Process Flow Diagram*
- *Use Case Diagram*
- *Activity Diagram*
- *Business Requirements Documentation*

21. Tools and Technologies Used

Tools Used

- *Microsoft Excel / Google Sheets*
- *Pivot Tables*
- *Data Cleaning Techniques*
- *Data Visualization*

- *Business Process Modeling*
- *Draw.io / Lucidchart*

Techniques Used

- *Data Analysis*
- *Trend Analysis*
- *Correlation Analysis*
- *KPI Development*
- *Business Requirement Analysis*
- *Process Flow Design*
- *Stakeholder-Oriented Reporting*

22. Conclusion

The Electric Vehicle Sales and Market Demand Analysis project demonstrates the practical application of business analysis and data analysis techniques to solve real-world business problems. The project successfully analyzed EV sales performance, charging infrastructure availability, fuel price trends, and market demand indicators to identify patterns and relationships affecting electric vehicle adoption.

Through structured data analysis, dashboards, and business process documentation, the project provided meaningful insights into sales growth trends, infrastructure impact, and market behavior. The findings highlight the importance of charging infrastructure development, fuel price monitoring, and demand trend analysis in driving EV adoption and business growth.

Overall, this project showcases the ability to translate raw data into actionable business insights and support strategic decision-making using business analysis methodologies.

.